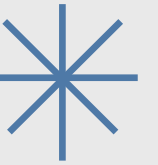


2025 DOMESTIC FLIGHT ANALYSIS



Ava McCann

BUSINESS PROBLEM



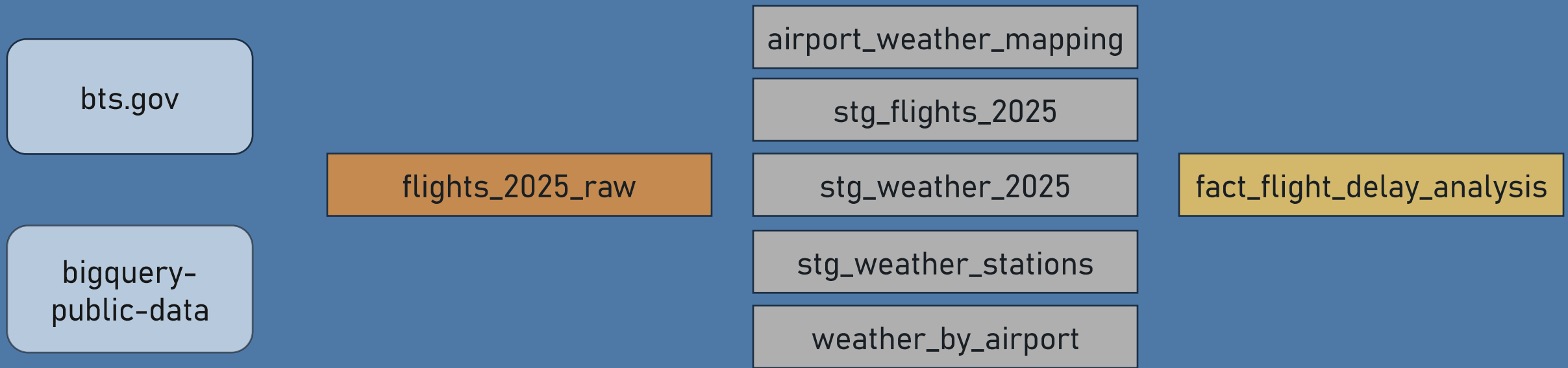
Airline delays and cancellations create operational inefficiencies, and customer dissatisfaction and increase costs. Understanding when and where disruptions occur can help identify high-risk areas and improve airline reliability.

North Star Metric:

On-Time Rate

Key Questions:

- Do certain airlines and airports operate more efficiently?
- What factors are top contributors to delays? Are delays and flight volume correlated?
- Are certain airports and routes higher risk?



METHODS

- Extracted 2025 domestic flight data from Bureau of Transportation Statistics.
- Extracted 2025 weather, weather station and airport data from BigQuery public data
- Implemented Medallion Architecture to clean, organize and structure data.
 - Mapped weather stations to the nearest airport to connect weather and flight data.
 - Integrated all flight, weather and airport data into OBT as a final output
- Final dashboard visualization created in Tableau

2025 Domestic Flight Data

Airports

Airlines

All
Airports

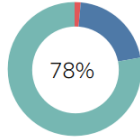
TOTAL FLIGHTS
7,001,618



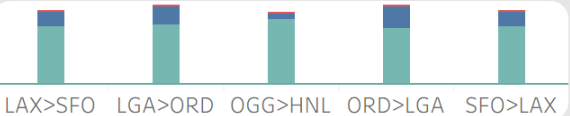
AVG DELAY
13.56



ON-TIME RATE

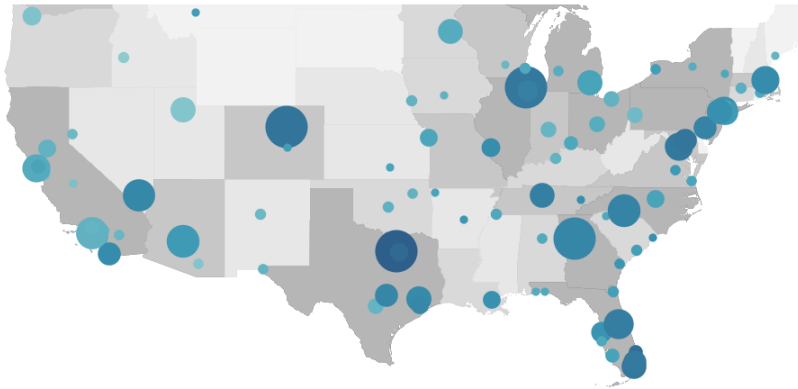


TOTAL ROUTES
6,938

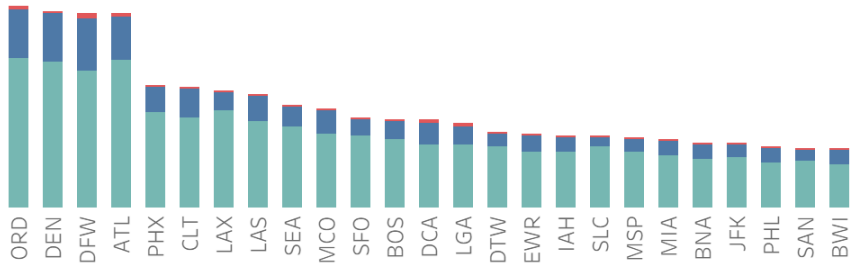


Delay Rate
0.1078 0.2776 All

Month of FI Date

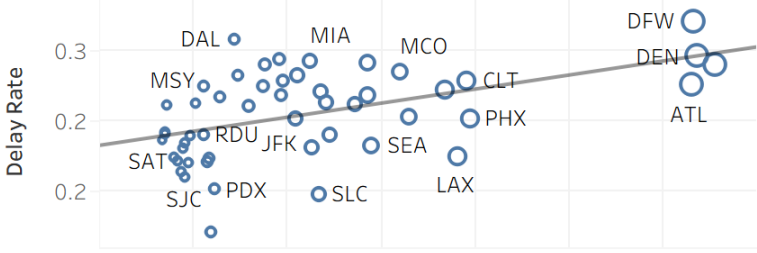
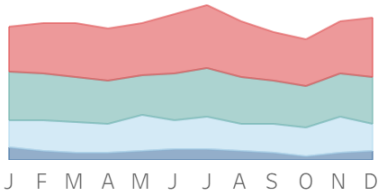
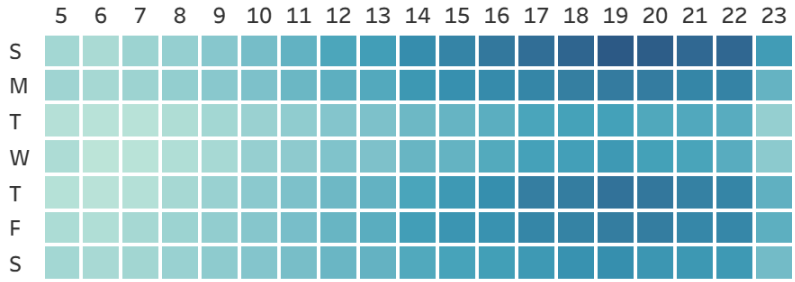


© 2026 Mapbox © OpenStreetMap



Average Delay by Weekday & Hour

Average Delay



INSIGHTS

Delays are highly seasonal

- Delay rates & times raise during summer months and winter holidays, loosely correlated with total flight volume.

Delays are correlated with busier airports

- Airport volume and delay rate show a positive relationship.

Delays compound throughout the day

- Flight volume stays steady while delays get longer, especially from 4-10pm. Low flight volume from 12am-5am allows for the delay time to be reset.

2025 Domestic Flight Data

Airports

Airlines

Delta

TOTAL FLIGHTS

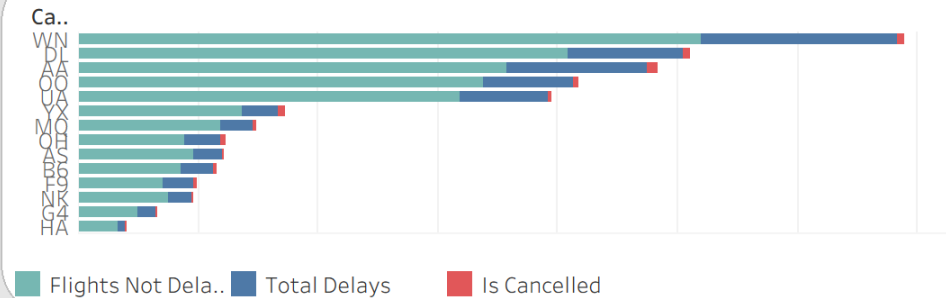
1,026,332

TOTAL ROUTES

1,167

ON-TIME RATE

80%



Filters

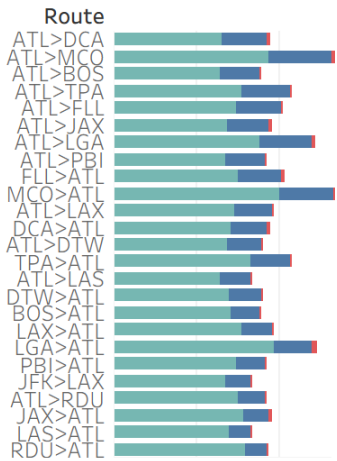
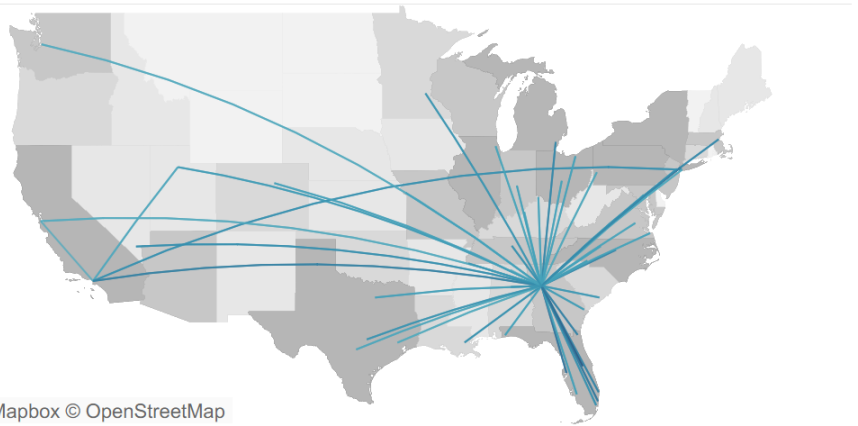
Carrier Code

DL

Month

All

Total Flights 0 5,375



Jan 12 Feb 9 Mar 9 Apr 6 May 4 Jun 1 Jun 29 Jul 27 Aug 24 Sep 21 Oct 19 Nov 16 Dec 14

Week [2025]

INSIGHTS

Top 5 airports dominate flight volume

Airline flight volume is not a strong predictor of delays

- Top airlines have more routes and more flights, but their reliability varies and does not correlate with their volume.

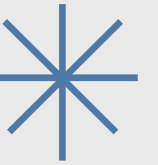
Route popularity is not a strong predictor of delays

- Top routes have widely varying delay rates with little correlation to their volume for the airline.

Delay risk for routes is related to airports

- Delays correlate the strongest with specific airports, especially ones with higher volumes and other high-risk airports.

CONCLUSION



Final Insights

- Flight delays and cancellations exhibit seasonal and hourly patterns.
- Airport congestion and certain bottleneck airports cause higher delay rates.
- Airline size and route volume alone are not reliable predictors of on-time performance, and it is more strongly correlated with airport volume.

Further Questions

To what extent are airline delay rates driven by the airports & routes they serve versus airline operational inefficiency?

Next Steps

- Develop predictive models for delay rates & cancellations
- Incorporate additional & current years of data through an automated data pipeline